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# Offshore & Deepwater Drilling

From Industry Fundamentals to Offshore Operations, Cost Control and HSE in Challenging Environments

www.londonpetroacademy.co.uk

# Offshore & Deepwater Drilling

This intensive 3-day Offshore & Deepwater Drilling course provides a practical, workshop-style introduction to how offshore wells are planned and drilled, and how these operations differ from land drilling. It starts with the big picture of the oil and gas industry and fundamentals of drilling, then moves into the specific technologies, rig types and operating constraints associated with shallow, deep and ultra-deepwater environments. Participants are guided through offshore drilling operations, directional and horizontal drilling, cost estimation and performance, and the logistics required to support operations in often harsh, remote locations. The program also addresses weather, marine support, sea port facilities and the use of floating units, subsea BOP stacks and marine risers, tying these elements back to drilling risk, performance and project economics. Health, safety, environmental and security issues are woven throughout, with case studies and scenario-based discussions to help attendees connect concepts directly to day-to-day offshore project decisions.

**LOCATION**

London

Houston

Dubai

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# COURSE OBJECTIVES

By the end of this course, participants will understand the fundamentals of offshore drilling and clearly explain how it differs from land drilling in terms of rig types, water depth, well design and logistics. They will be able to describe the key components of offshore drilling rigs, the basics of well design, drilling bits, directional and horizontal drilling, and routine drilling operations including monitoring and well control. Attendees will learn how to perform drilling cost analysis, estimate drilling costs, calculate daily rig rates, and distinguish between fixed and variable costs, contingencies and non-productive time. They will be able to explain the reasons for directional drilling, identify the main tools and measurements involved, and apply principles of drilling hydraulics, bit selection and optimisation to improve rate of penetration and cuttings transport. Participants will also understand the role of weather, supply vessels, helicopters and shore bases in offshore logistics, and appreciate key health, safety, security and environmental considerations, including regulatory frameworks, oil spill prevention and emergency response in the offshore drilling environment.

This course is designed for oil and gas professionals involved in, or transitioning into, offshore drilling projects who need a strong, integrated understanding of offshore and deepwater operations. It is particularly suitable for drilling engineers, drilling superintendents and foremen, reservoir engineers, petrophysicists, geoscientists, economists and planners who must evaluate or support offshore drilling campaigns. Facilities planning engineers, accountants, mid-level management and other technical staff who influence offshore project decisions and budgets will also benefit from the commercial and operational insights provided. The course is appropriate for both early-career professionals seeking a structured grounding in offshore drilling and experienced land-drilling personnel who want to broaden their expertise into offshore and deepwater environments.

# Day 1

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## Big Picture of the Oil & Gas Industry

- Meaning of petroleum
- Typical oil and gas company objectives
- Industry streams
- World reserves and production
- Peak oil
- Production management
- Quotas and capacities (OPEC/Non-OPEC)
- Market distribution and dynamics
- Role of IOCs, NOCs and regulatory bodies
- Glossary of terms

## Fundamentals of Drilling

- The drilling rig: types and components
- The drilling team
- Drilling fluids (mud) and circulating system
- Basic well design
- Drilling bits
- Directional and horizontal drilling
- Routine drilling operations
- Well monitoring
- Well control
- Wellbore problems and preventions
- Special drilling operations (coring, fishing, etc.)

Case Studies: Mud weight window, rig horse-power and drilling depth

## Offshore Drilling

- Differences between land and offshore
- Water depth and rig types (deep water MODUs – Mobile Offshore Drilling Units)
- Sea bed preparation
- Fixed platform
- Floating drilling and station keeping
- Motion compensation
- Conductor casing (jetting/riserless drilling)
- Subsea BOP stack
- Marine/production riser for various deep water applications
- Slip joint
- Rotating head
- ROVs

# Day 2

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**Drilling Cost Analysis**

- Drilling cost estimation
- Authorisation for Expenditure (AFE)
- Daily rig rate
- Fixed operating costs
- Variable costs
- Drilling contingencies
- Non-productive time
- Drilling performance and optimization

Case Studies: Cost per foot, bit performance

**Directional/Horizontal Drilling**

- Reasons for directional drilling
- Definitions
- Directional tools
- Well trajectories
- Directional drilling measurements
- Hole cleaning
- Extended reach wells (case study)
- Drilling Bits
- Types of bits
- Rock failure mechanisms
- Bit selection and evaluation
- Factors affecting rate of penetration

**Drilling Hydraulics**

- Hydrostatic pressure
- Buoyancy
- Rheological models
- Bit nozzle size selection
- Drilling hydraulic optimization
- Hole cleaning/cutting transport

# Day 3

- Drilling Fluids (Mud)
- Functions of drilling fluids
- Mud properties
- Water-based muds
- Oil-based muds
- Logistic Support and Services
- Weather conditions
- Supply vessels
- Helicopter
- Land base
- Sea port facility
- Health, Safety, Environment and Security
- Health, safety, environment and security
- Elements of drilling/production safety and regulations
- Think of unthinkable (scenario planning)
- Minimal operational requirements
- Learning from disasters
- Oil spill prevention and response
- First responders and emergency equipment
- U-Turn: work through your own problems and walk away with real solutions to your workplace challenges!
- Well Completion and Production
- Near wellbore formation damage
- Evaluating a well, logging, MWD and LWD
- Types of completions
- Perforating a well
- Well testing
- Reservoir stimulation
- Completion equipment, concepts and techniques
- Multizone completions
- Artificial lift technique
- Workover operations
- DW field development costs
- FPSO/subsea schemes instead of floating platforms
- Case Studies
- Course Summary and Wrap-Up

# Contact us


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Registered in England & Wales No. 09875694
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